APPLIED DATA SCIENCE CAPSTONE

Predicting Falcon 9 First-Stage Landing Success

A data science analysis of SpaceX launch outcomes — from data collection through predictive modeling

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Github: https://github.com/ptah222/IBM-Data-Science-Capstone-Project.git

OVERVIEW

Executive Summary

- This project aims to predict whether the Falcon 9 first stage will land successfully so that a competing company can make more informed launch bids.
- Methods used: data collection (API + web scraping), data wrangling, exploratory data analysis, interactive visual analytics and predictive classification modeling.
- After comparing multiple models, we find that the Falcon 9 booster has a high probability of successful landing (87.7% based on decision tree classifier). Success is highly correlated with launch site, orbits, boosters and payload mass.
- Success rate has reached 90% in recent years, thanks to new generation of boosters.

87.7%

Best model accuracy
Decision Tree classifier

90

Launch records analyzed
Falcon 9 launches (2010–2020)

66.7%

Overall landing success rate
60 of 90 landings succeeded

BACKGROUND

Introduction

Project Background

- SpaceX advertises Falcon 9 launches at ~$62M, far below competitors' ~$165M+, largely because SpaceX reuses the first stage instead of discarding it.
- Reuse is only possible when the first stage successfully lands, so landing success directly determines launch cost.
- A competing company bidding against SpaceX would benefit from estimating this cost — which requires predicting landing success.

Problem Statement

Can we predict whether the Falcon 9 first stage will land successfully, using publicly available launch data — payload mass, orbit type, launch site, booster version, and more?

Key Questions Explored

- Which variables most influence landing outcome?
- How has success rate changed over time?
- Which classification model predicts outcome best?

DATA COLLECTION

SpaceX REST API

1 GET request to api.spacexdata.com/v4/launches/past

2 Parse JSON response into a pandas DataFrame

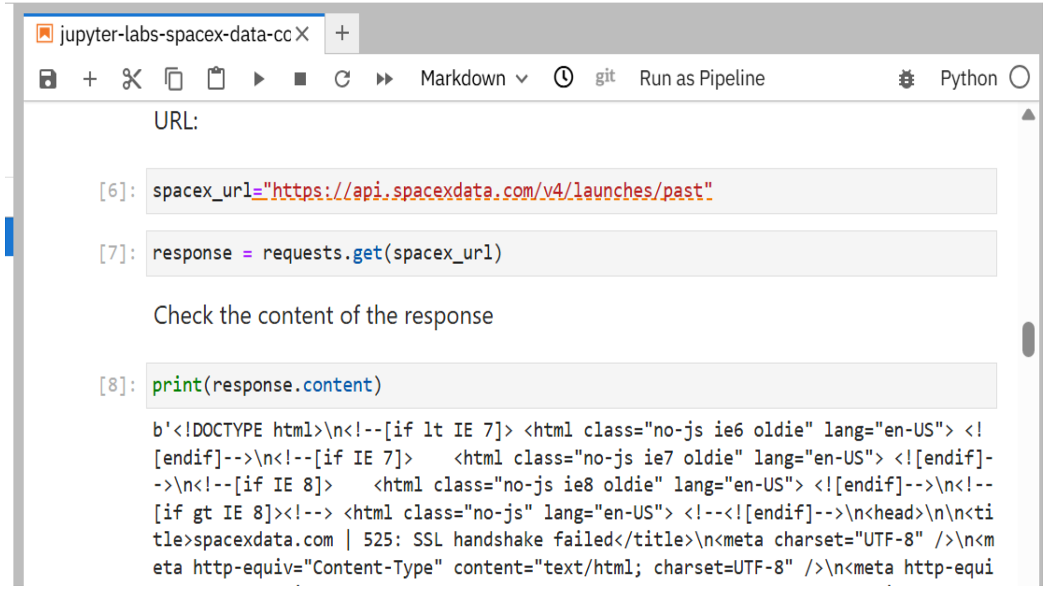
3 Use rocket, payload, launchpad, core IDs to make follow-up GET requests for details

4 Merge nested results into flat columns per launch

5 Filter to Falcon 9 launches only (drop Falcon 1)

Key Steps

- requests.get(url).json() against the SpaceX v4 API
- IDs returned per launch (rocket, payloads, launchpad, cores) require separate GET calls to resolve into readable fields
- Custom functions (e.g. getBoosterVersion(), getLaunchSite()) map IDs → names and append to lists
- Combine into a single DataFrame; export to spacex_api.csv



Notebook / Code: *github.com/ptah222/IBM-Data-Science-Capstone-Project*

DATA COLLECTION

Web Scraping (Wikipedia)

1 GET the Wikipedia “List of Falcon 9 and Falcon Heavy launches” page

2 Parse HTML with BeautifulSoup

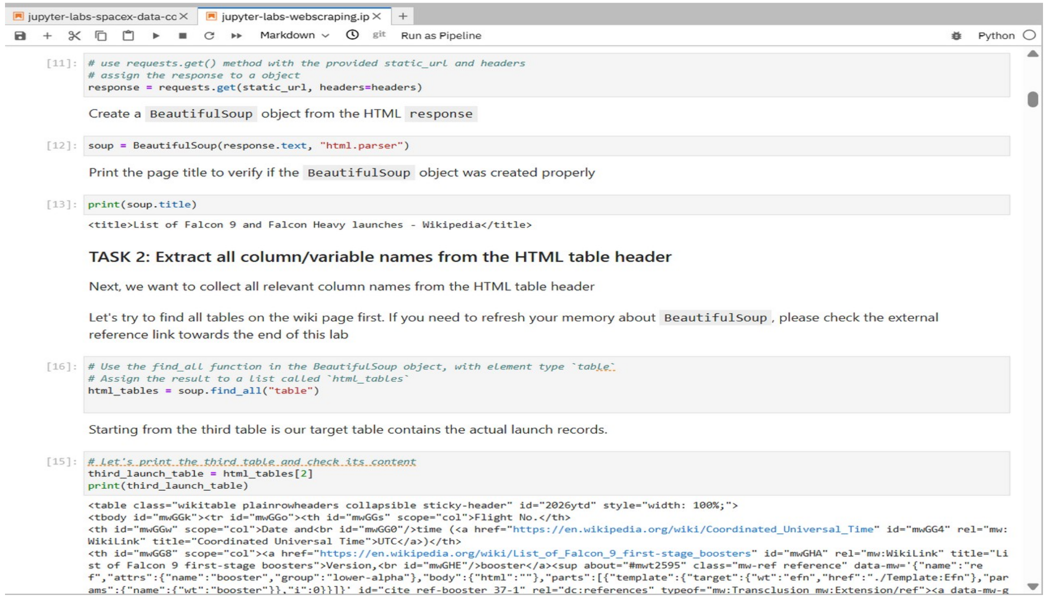
3 Locate the launch-record HTML table

4 Extract each column (date, booster, site, payload, outcome)

5 Assemble rows into a DataFrame

Key Steps

- requests.get() the Wikipedia revision URL, then BeautifulSoup(html, 'html.parser')
- soup.find_all('table', 'wikitable plainrowheaders collapsible') to locate the launch table
- Custom parsing functions per column (date/time, booster version, payload mass, orbit, outcome)
- Append parsed cells into a dict of lists → pd.DataFrame; export to spacex_web_scraped.csv



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PREPARATION

Data Wrangling Methodology

1 Combine API + scraped datasets; resolve column mismatches

2 Handle missing values (e.g. impute mean payload mass)

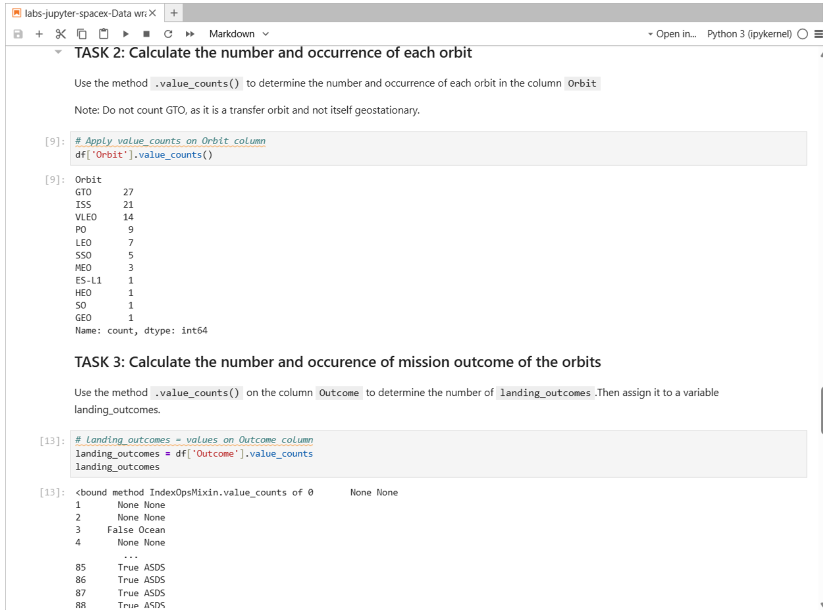
3 Standardize categorical labels (site, orbit, outcome text)

4 Engineer the binary Class label (1 = success, 0 = failure) from landing outcome text

5 Export the clean, model-ready dataset

What Counts as a "Successful Landing"

- Outcome strings like 'True ASDS', 'True RTLS', 'True Ocean' → Class = 1
- 'False ASDS', 'False RTLS', 'None None' → Class = 0
- This binary label becomes the target variable for the classification models



Notebook / Code: *github.com/ptah222/IBM-Data-Science-Capstone-Project*

EXPLORATORY ANALYSIS

EDA with Data Visualization

Charts Used & Purpose

Scatter — Flight Number vs. Launch Site	Reveal whether success improved over successive flights, per site
Scatter — Payload Mass vs. Launch Site	Check whether heavier payloads cluster at certain sites or fail more
Bar — Success Rate by Orbit Type	Compare landing reliability across orbit classes (LEO, GTO, ISS, …)
Scatter — Flight Number / Payload vs. Orbit Type	Look for interaction effects between orbit, payload, and experience
Line — Yearly Launch Success Trend	Show how success rate evolved as SpaceX iterated on booster design

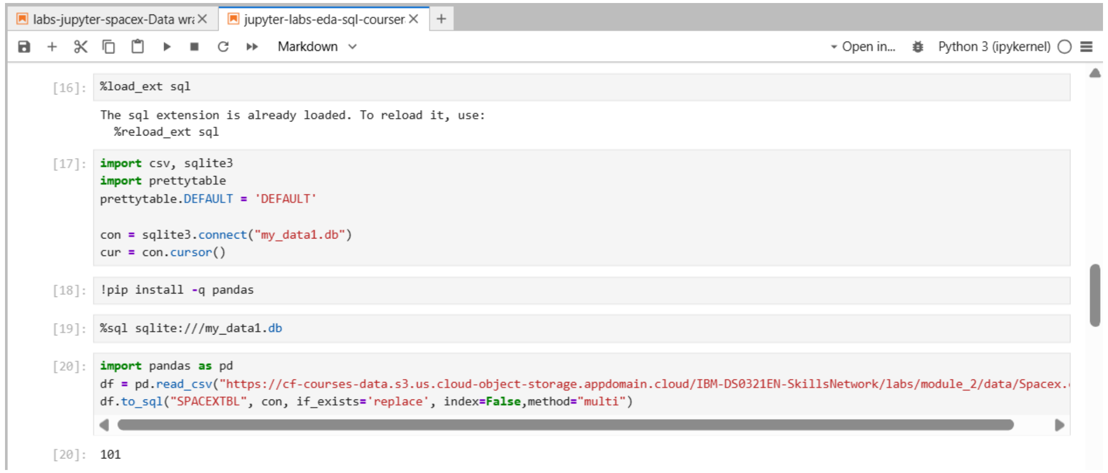
- Each chart tests one hypothesis about landing success — experience, mass, orbit energy, or time — and every one of them fed a feature into the final classifier.

Notebook / Code: *github.com/ptah222/IBM-Data-Science-Capstone-Project*

EXPLORATORY ANALYSIS

EDA with SQL

- Loaded the wrangled dataset into a SQLite / Db2 table via %sql / ibm_db
- Queries covered: distinct launch sites, payload totals by customer, average payload by booster, first successful landing date, boosters meeting payload + outcome criteria, mission-outcome counts, max-payload boosters, month-level 2015 records, and a ranked landing-outcome count (2010–2020)
- Each query answers one specific analytical question used later in the results section



Query Categories Covered

- Launch Sites
- Payloads
- Success Rates
- Rankings
- Time Analysis

Notebook / Code: *github.com/ptah222/IBM-Data-Science-Capstone-Project*

EXPLORATORY ANALYSIS

Interactive Visual Analytics

Folium Map

- Marker for every launch site, colored by success/failure ratio
- Circle markers for each individual launch record, clustered by site
- Distance lines from a launch site to nearby coastline, highway, and city, to explore siting rationale

Why it matters

- The map makes the siting logic visible: every pad is coastal, so a failed first stage falls over open water rather than populated land.

Plotly Dash App

- Dropdown to filter by launch site
- Pie chart of success-rate share across sites (or per-site success vs. failure)
- Range slider + scatter chart of payload mass vs. success, colored by booster version

Why it matters

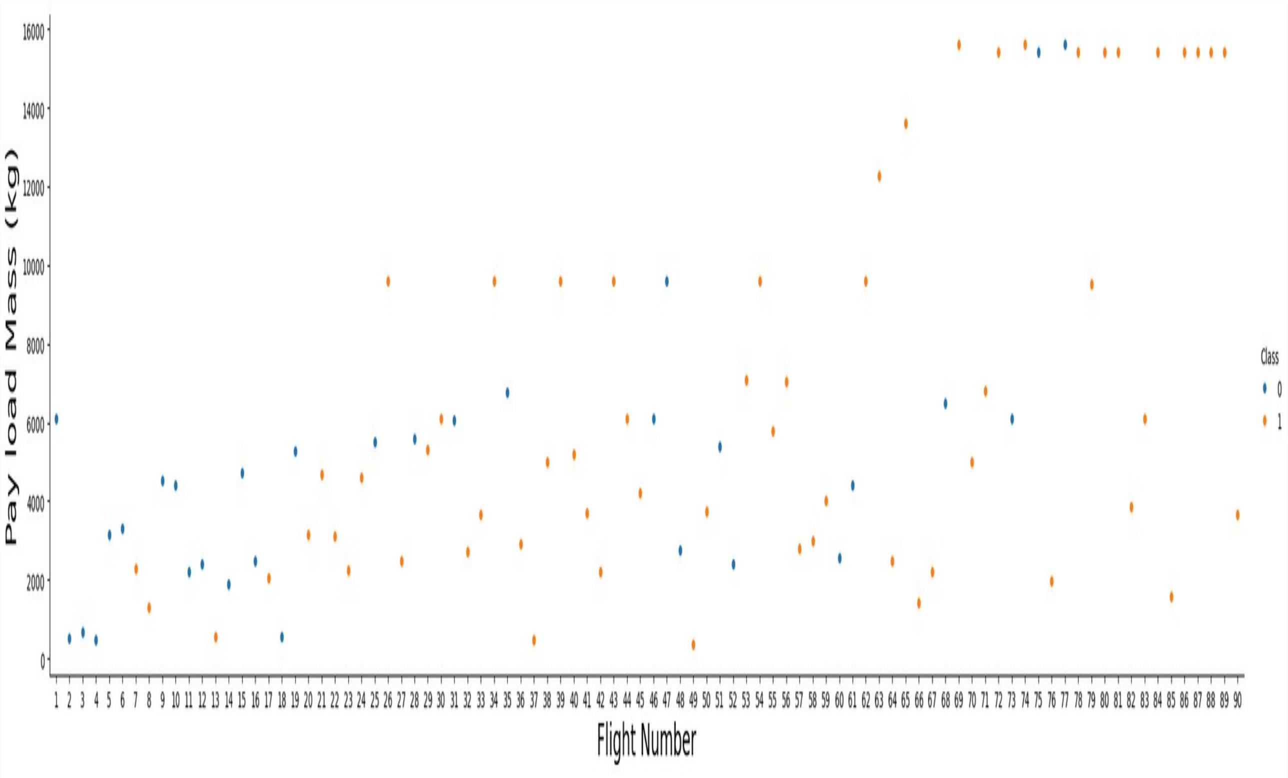
- The dashboard lets a non-technical bidder test payload and site combinations directly, without rerunning any of the notebooks.

Notebook / App Code: *github.com/ptah222/IBM-Data-Science-Capstone-Project*

EDA RESULTS

Flight Number & Payload vs. Launch Outcome

Each point is one Falcon 9 launch; orange = successful landing (Class 1), blue = failure (Class 0)



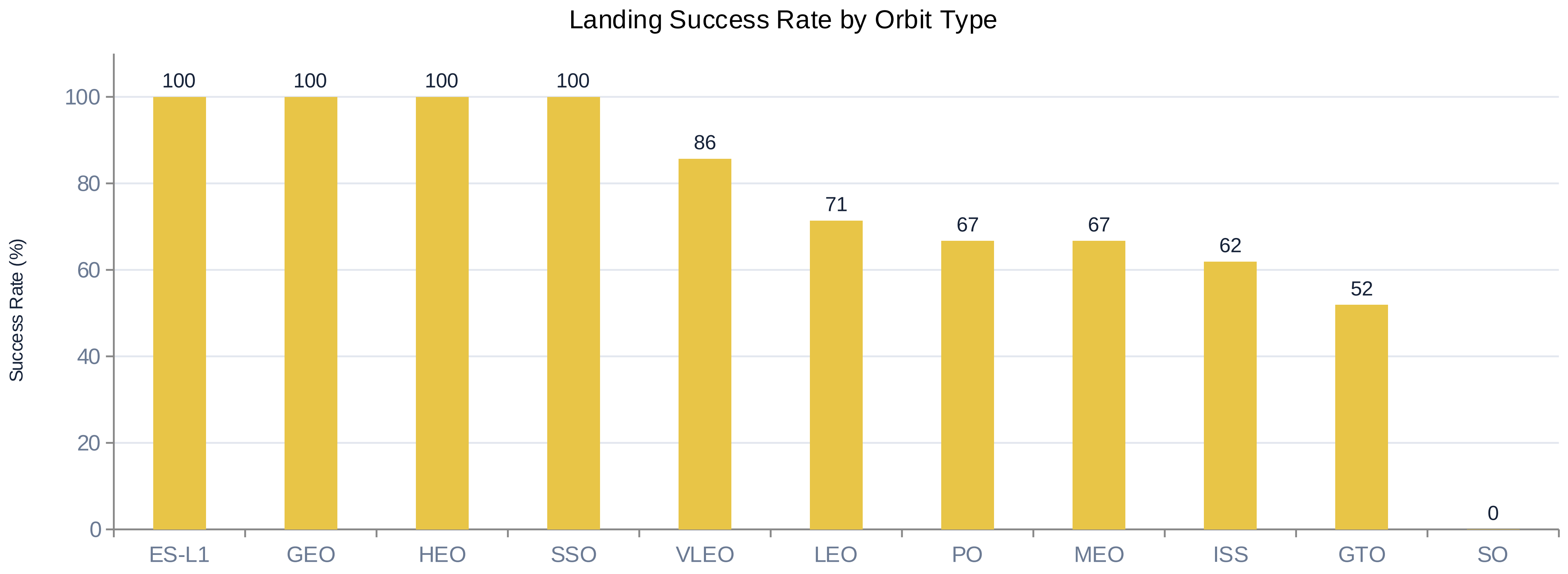
Flight Number vs. Payload Mass, colored by outcome (Class)

EDA Notebook: *github.com/ptah222/IBM-Data-Science-Capstone-Project*

EDA RESULTS

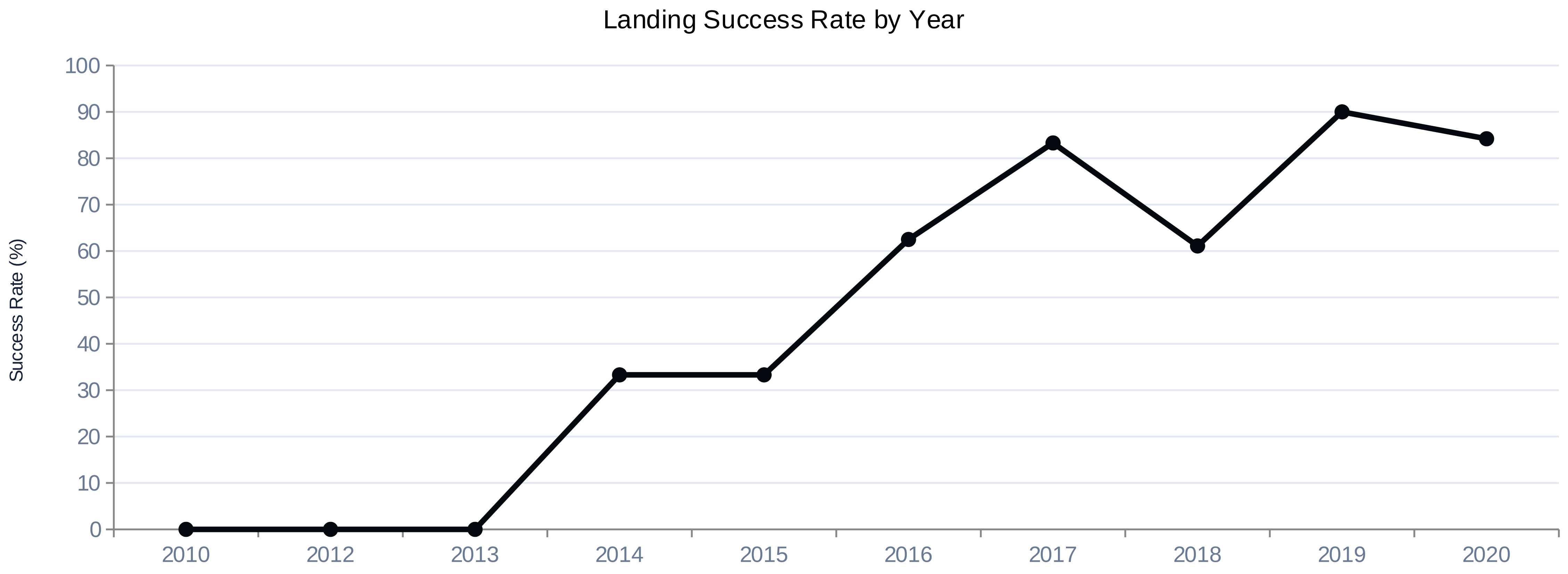
Success Rate by Orbit Type

ES-L1, GEO, HEO and SSO orbits show a 100% landing success rate; GTO is lowest at 51.9%, reflecting its higher energy requirements.



Yearly Launch Success Trend

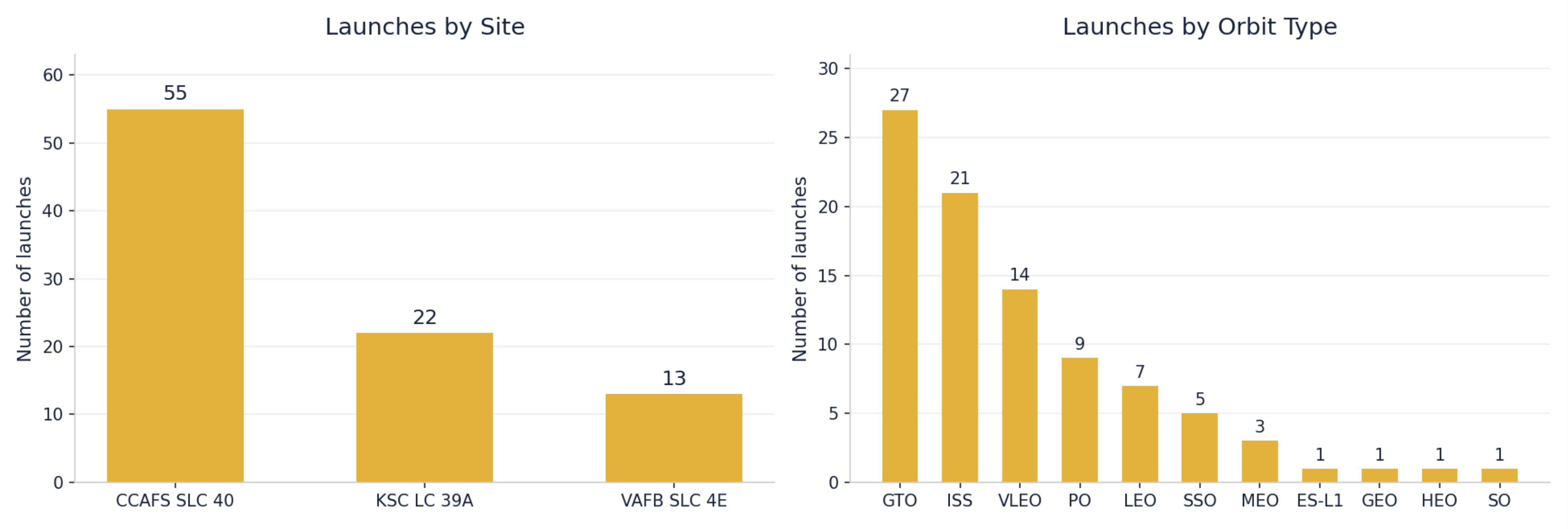
Success rate climbed from 0% before 2014 to 90% in 2019 as SpaceX iterated on Block 5 boosters and landing technique.



EDA RESULTS

Launch Site & Orbit Distribution

Counts computed with value_counts() over the 90-launch wrangled dataset



- CCAFS SLC 40 handled 55 of 90 launches; GTO and ISS together account for 48, so the dataset is dominated by one pad and two orbit classes — worth remembering when reading per-site and per-orbit success rates.

Both charts total 90 launches

EDA Notebook: *github.com/ptah222/IBM-Data-Science-Capstone-Project*

Launch Sites & Payload Totals

Query	Sample Result
Distinct launch site names	CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E
Total payload mass carried for customer NASA (CRS)	45,596 kg

What these queries establish

- The SQL table lists four pad identifiers, but CCAFS LC-40 and CCAFS SLC-40 are the same pad renamed in 2016 — the wrangled dataset collapses them to three sites.
- NASA (CRS) resupply missions account for 45,596 kg of payload in total, making NASA the single largest customer by mass.
- The early F9 v1.1 booster averaged only 2,928.4 kg per launch — later Block 5 boosters carry several times that, so payload mass and booster generation move together.

Results copied from the executed EDA-SQL notebook (Spacex.csv loaded into SPACEXTABLE).

EDA RESULTS — SQL

Success Rates & Rankings

Query	Sample Result
First successful ground-pad landing date	2015-12-22
Boosters: drone-ship success & payload 4,000–6,000 kg	F9 FT B1022, F9 FT B1026, F9 FT B1021.2, F9 FT B1031.2
Mission outcome counts (all records)	Success: 100 | Failure (in flight): 1
Boosters carrying the maximum payload mass	12 F9 B5 boosters — B1048.4, B1049.4, B1051.3, B1056.4, ...

Results copied from the executed EDA-SQL notebook (Spacex.csv loaded into SPACEXTABLE).

Time Analysis

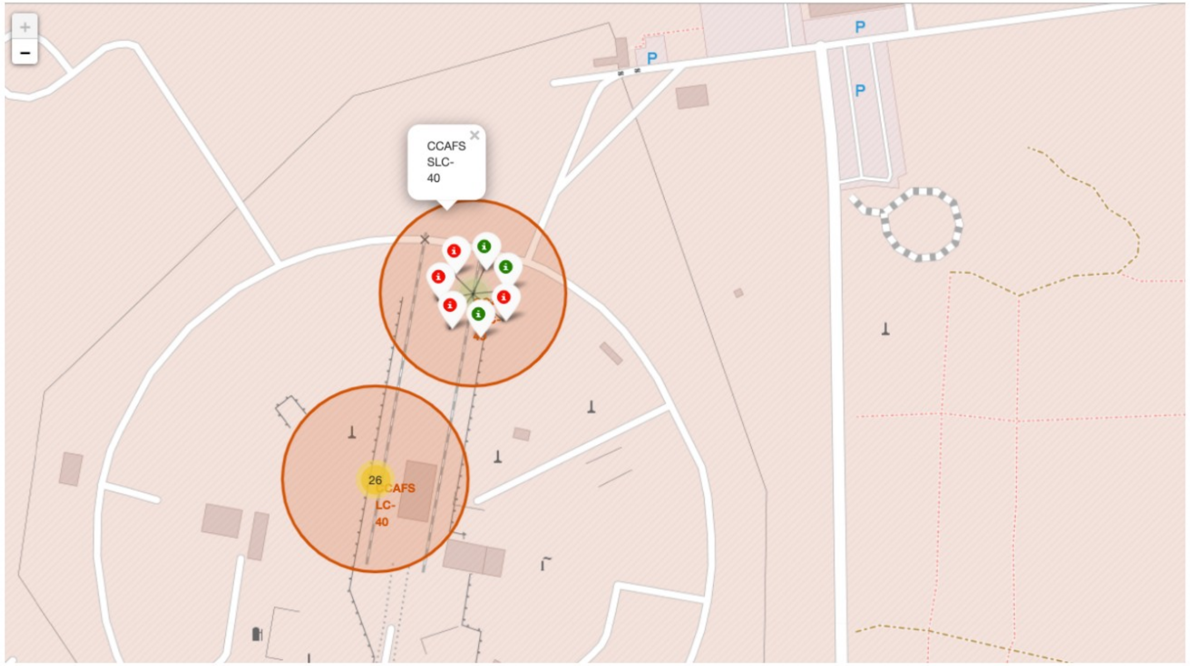
Query	Sample Result
2015 drone-ship landing failures, by month	January and April (F9 v1.1 B1012 and B1015, both CCAFS LC-40)
Ranked landing outcomes, 2010-06-04 to 2017-03-20	1) No attempt — 10 2) Success drone ship — 5 3) Failure drone ship — 5 4) Success ground pad — 3

What the time analysis shows

- Both 2015 drone-ship failures came in the first half of the year, on F9 v1.1 boosters flying from CCAFS LC-40 — the period when barge landing was still unproven.
- Through March 2017 the most common outcome was still no recovery attempt (10), with drone-ship attempts split evenly at 5 successes and 5 failures.
- Ground-pad returns were rarer but more reliable (3 successes, no failures), which is why SpaceX kept investing in both recovery modes.

Results copied from the executed EDA-SQL notebook (Spacex.csv loaded into SPACEXTABLE).

Folium Map — Launch Site Markers



From the color-labeled markers in marker clusters, you should be able to easily identify which launch sites have relatively high success rates.

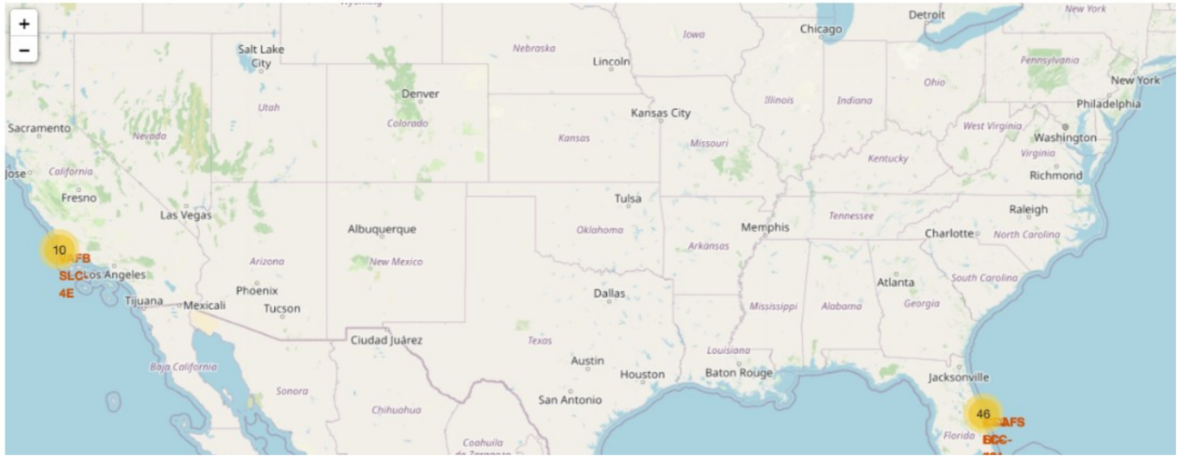
Site markers — folium.Circle + folium.Marker

```
nasa_coordinate = [29.5597, -95.0831]
site_map = folium.Map(
    location=nasa_coordinate, zoom_start=5)

for _, row in launch_sites.iterrows():
    coord = [row['Latitude'], row['Longitude']]
    folium.Circle(
        coord, radius=1000, color='#000000',
        fill=True
    ).add_child(
        folium.Popup(row['LaunchSite'])
    ).add_to(site_map)
    folium.Marker(
        coord, popup=row['LaunchSite']
    ).add_to(site_map)
```

Folium Notebook: *github.com/ptah222/IBM-Data-Science-Capstone-Project*

Folium Map — Launch Records & Proximities



All 90 launch records, color-coded by outcome and clustered by site

Marker clusters & proximity analysis

- Every launch record is added to a MarkerCluster, with marker_color set from the Class label — green for a successful landing, red for a failure.
- Clusters expand on zoom, so overlapping launches at CCAFS SLC-40, KSC LC-39A and VAFB SLC-4E stay readable.
- folium.PolyLine plus a Haversine distance calculation measure each site to its nearest coastline, railway, highway and city.

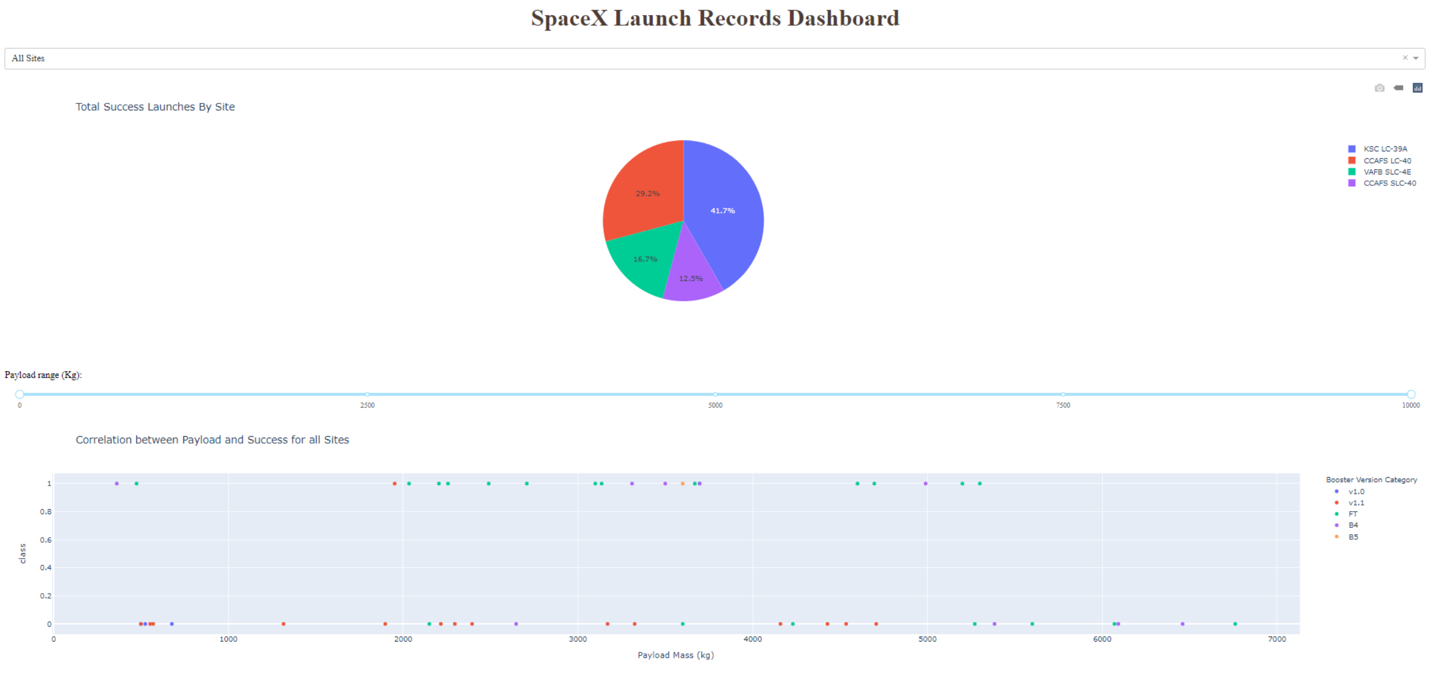
What the map shows

- All three sites sit within roughly 1 km of the coast — failed stages fall over water — while cities are kept much further away.
- CCAFS SLC-40 and KSC LC-39A carry the bulk of launches and show visibly denser green clusters than VAFB SLC-4E.

Folium Notebook: *github.com/ptah222/IBM-Data-Science-Capstone-Project*

INTERACTIVE VISUAL ANALYTICS

Plotly Dash — Interactive Dashboard



Success pie chart

- With All Sites selected, the pie splits total successful launches by site: KSC LC-39A leads at 41.7%, followed by CCAFS LC-40 at 29.2%.
- Selecting a single site switches the chart to that site's own success vs. failure share.

Payload vs. outcome scatter

- Points are colored by booster version category, and the payload range slider filters the axis.
- Successful landings concentrate below roughly 6,000 kg; FT and B4 boosters account for most of the successes.

Dash App Code: *github.com/ptah222/IBM-Data-Science-Capstone-Project*

PREDICTIVE MODELING

Predictive Analysis Methodology

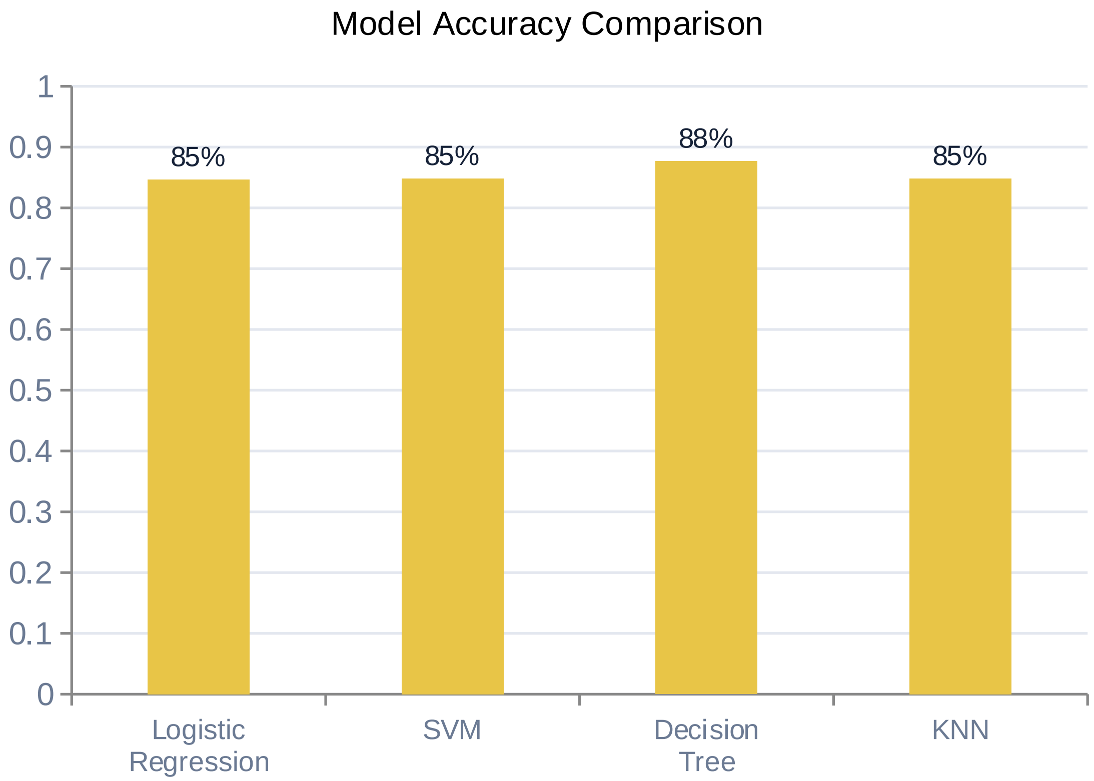
1 Standardize numeric features (StandardScaler)	**2** Train/test split of the labeled dataset	**3** Grid-search hyperparameters with cross-validation, per model	**4** Fit Logistic Regression, SVM, Decision Tree, and KNN classifiers	**5** Compare test-set accuracy; select the best model

- Target variable: Class (1 = successful landing, 0 = failure), engineered during data wrangling
- Features: flight number, payload mass, orbit type, launch site, booster version/grid fins/legs, and reused-booster flag
- Each model tuned via GridSearchCV; best estimator per model type evaluated on the held-out test set

Modeling Notebook: *github.com/ptah222/IBM-Data-Science-Capstone-Project*

PREDICTIVE MODELING

Model Comparison & Best-Model Results



Best Model: Decision Tree — 87.7% Accuracy

Best params: gini, max_depth=8, max_features=sqrt, min_samples_leaf=4, min_samples_split=2

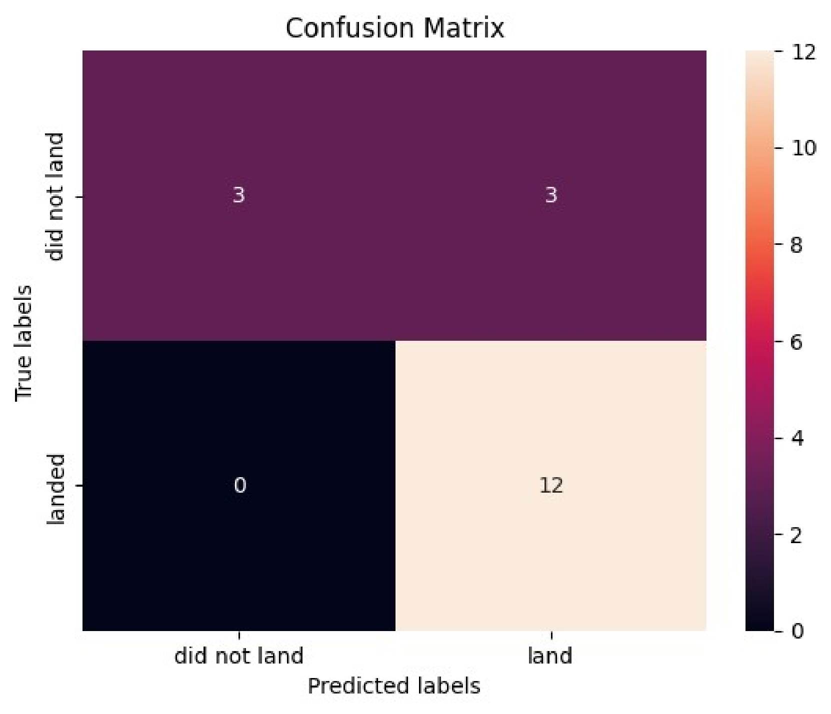
- Tuned with GridSearchCV over a 5-fold cross-validation split.
- All four models exceed 84%, so the signal is in the features, not the algorithm.
- All four models produce an identical confusion matrix and the same 83.3% test accuracy — the CV score is what separates them.

Decision Tree scored best in cross-validation (87.7%), ahead of SVM/KNN (84.8%) and Logistic Regression (84.6%), and was selected as the final model.

PREDICTIVE MODELING

Confusion Matrix — Decision Tree

Held-out test set of 18 launches; identical for all four classifiers



Reading the matrix

- True positives: 12 — every launch that actually landed was predicted to land.
- False negatives: 0 — the model never misses a successful landing.
- True negatives: 3, false positives: 3 — of six real failures, half were wrongly predicted as successes.
- Accuracy = 15/18 = 83.3% on the test set.

Why it matters

- The single weakness is false positives. For a competitor bidding against SpaceX, that means the model may assume a cheap reusable launch when the stage would in fact be lost — the costly direction to be wrong in.

Rows = true outcome, columns = predicted outcome

Modeling Notebook: *github.com/ptah222/IBM-Data-Science-Capstone-Project*

Conclusion

- The Decision Tree Classifier was the best performing model, at 87.7% cross-validated accuracy — ahead of SVM and KNN (84.8%) and Logistic Regression (84.6%).
- The strongest predictors of a successful landing are flight number (accumulated experience), orbit type, launch site and payload mass.
- Landing success is strongly time-dependent: 0% before 2014, rising to 90% by 2019 as SpaceX iterated toward the Block 5 booster.
- Orbit type matters sharply — ES-L1, GEO, HEO and SSO reached 100% success, while high-energy GTO missions managed only 51.9%.
- KSC LC-39A had the highest per-site success rate; heavier payloads above ~7,000 kg showed noticeably fewer successful landings.
- A competitor can therefore estimate whether SpaceX will recover a first stage — and so whether a bid is priced against a ~$62M reusable launch or a full-cost expendable one.

GitHub Repository Links

- Data Collection — API notebook: https://github.com/ptah222/IBM-Data-Science-Capstone-Project/blob/0584d6d27082113542dc9254eb112fbda10bb4f8/part%201%20jupyter-labs-spacex-data-collection-api.ipynb
- Data Collection — Web Scraping notebook: https://github.com/ptah222/IBM-Data-Science-Capstone-Project/blob/bb0d5b67337cbdf7ec640d6aaa9d6e046e613737/part%201%20jupyter-labs-webscraping.ipynb
- Data Wrangling notebook: https://github.com/ptah222/IBM-Data-Science-Capstone-Project/blob/03cf6dd686448d61155a02de20f39b63c7e72e58/labs-jupyter-spacex-Data%20wrangling.ipynb
- EDA — Data Visualization notebook: https://github.com/ptah222/IBM-Data-Science-Capstone-Project/blob/f95268ea8e1c61bacbb2be381b085205bc301cb0/edadataviz.ipynb
- EDA — SQL notebook: https://github.com/ptah222/IBM-Data-Science-Capstone-Project/blob/f95268ea8e1c61bacbb2be381b085205bc301cb0/part%202%20jupyter-labs-eda-sql-coursera_sqllite.ipynb
- Interactive Visual Analytics — Folium notebook: https://github.com/ptah222/IBM-Data-Science-Capstone-Project/blob/c4cfb8f28e87c02f9103472e11f476a3fd3c9572/lab_jupyter_launch_site_location_folium.ipynb
- Interactive Visual Analytics — Plotly Dash app: https://github.com/ptah222/IBM-Data-Science-Capstone-Project/blob/0a065d79f08a9dac8a9bd57f4a0756986d7ecb52/spacex-dash-app.py
- Predictive Analysis — Modeling notebook: https://github.com/ptah222/IBM-Data-Science-Capstone-Project/blob/2f4e7255690fa8ce7c7aa9f9847a3803ac14a9fd/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb